

PROJECT DESIGN PHASE

Electric Vehicle Analytics Dashboard using Flask & Tableau E-CarStart

Project Information

Project Name	Electric Vehicle Analytics Dashboard
Website	E-CarStart
Frontend	HTML, CSS, Bootstrap
Backend	Python Flask
Visualization	Tableau Public

Introduction

The Project Design Phase defines the overall architecture and workflow of the Electric Vehicle Analytics Dashboard. It explains how the Flask application, Tableau Public dashboards, and frontend components work together to provide an interactive analytics platform.

Existing System

EV information is often scattered across different reports and websites, making analysis difficult for users.

Proposed System

The proposed solution integrates Tableau Public dashboards into a Flask web application, allowing users to explore EV registrations, manufacturer trends, vehicle categories, and interactive stories through a responsive interface.

System Architecture

Presentation Layer: HTML, CSS, Bootstrap
Application Layer: Flask
Visualization Layer: Tableau Public
Data Layer: Electric Vehicle Registration Dataset

High-Level Design

- Home Module
- Dashboard Module
- Story Module
- About Project Module
- Navigation Module

Low-Level Design

The Home page provides project information and navigation. The Dashboard page embeds Tableau analytics. The Story page embeds Tableau Story. Flask routes handle page navigation and template rendering.

System Workflow

User → Flask Application → HTML Template → Tableau Dashboard/Story → Interactive Visualizations

Module Design

1. Home Module
2. Dashboard Module
3. Story Module
4. About Project Module

Technology Architecture

Frontend: HTML, CSS, Bootstrap

Backend: Flask

Visualization: Tableau Public

Version Control: Git & GitHub

Benefits of the Design

- Modular architecture
- Responsive interface
- Easy maintenance
- Interactive analytics
- Scalable design

Conclusion

The proposed design ensures an organized, scalable, and user-friendly analytics platform that effectively presents Electric Vehicle insights through Tableau integrated with Flask.